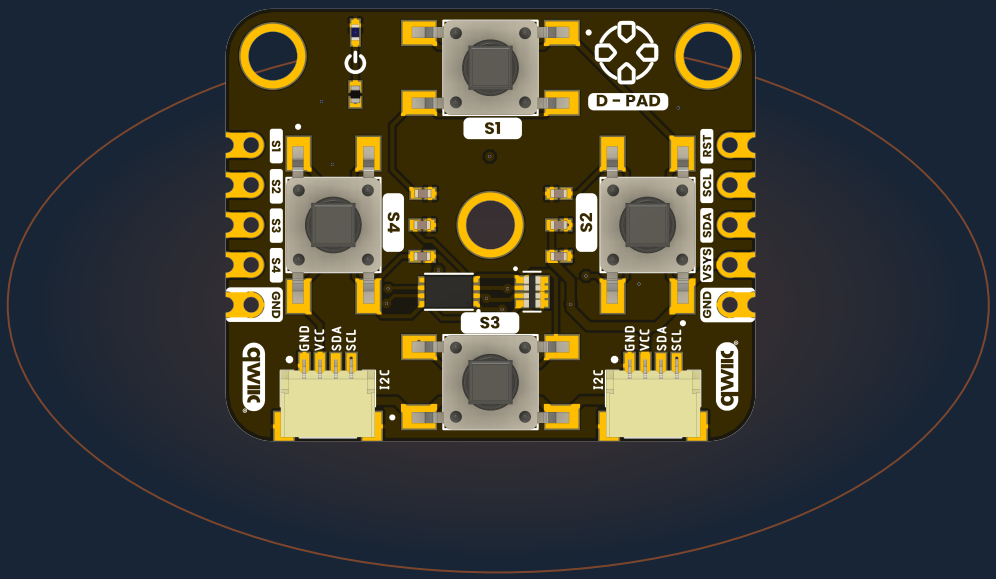
PRODUCT REFERENCE MANUAL

UNIT ELECTRONICS · DEVLAB ECOSYSTEM

DevLab I2C D-Pad Module

MPN: UE0173

Version 1.1.0



I2C D-Pad module with four directional controls.

D-PAD | I2C | DEVLAB ECOSYSTEM

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Description

The DevLab I2C D-Pad Module (MPN **UE0173**) is an I2C directional input module in the UNIT Electronics DevLab family. The product-specific board views and any released drawings are collected under `hardware/`.

Applications

- Directional input in an I2C-based prototype.
- Evaluation of the module with a compatible host controller.

Available Documentation

A product-specific schematic was not present in the selected source package. See Chapter 8 for the available source files. Confirm the electrical interface against the product-specific pinout or schematic before connecting the board.

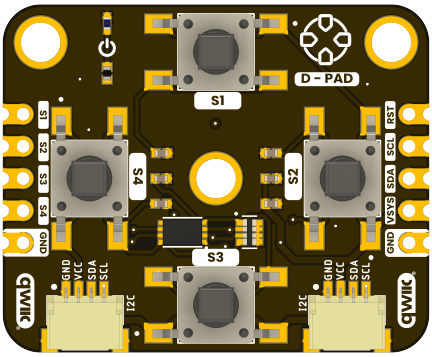
1 The Board

1.1 Board Identification

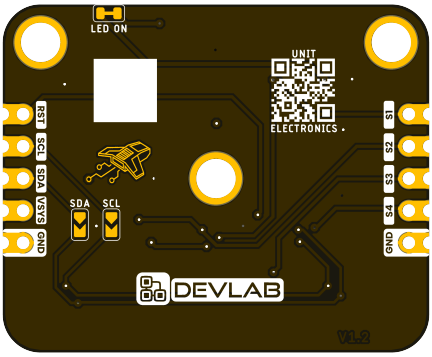
Item	Identification
Product	DevLab I2C D-Pad Module
Manufacturer Part Number (MPN)	UE0173
Function	Directional input

1.2 Board Views

The following views come from the selected UE0173 source package. They show component placement and board orientation; use a dimensioned drawing for mechanical tolerances.



DevLab I2C D-Pad Module top view.



DevLab I2C D-Pad Module bottom view.

2 Ratings

The selected UE0173 source package does not establish a complete set of electrical ratings in this reference. Do not use values from another DevLab module as substitutes.

2.1 Electrical Limits

Supply range, I/O voltage limits, and current consumption require confirmation from the product-specific schematic and component data before integration.

2.2 Interface Characteristics

Confirm the I2C address, bus speed, pull-up arrangement, and connector orientation with the released UE0173 pinout or firmware documentation. An I2C scan only identifies responding addresses; it does not verify the device protocol.

3 Functional Overview

The DevLab I2C D-Pad Module provides directional input for an I2C-based system. The supplied board views identify the physical module, while the product-specific schematic and firmware documentation are needed to establish its internal signal path and command set.

3.1 Integration Boundary

Use the available UE0173 pinout, schematic, and firmware release to determine power, connector mapping, address, and commands. Until these are confirmed, treat the generic I2C scanner in `software/` as a bus diagnostic only.

4 Connectors and Pinout

A released pinout drawing for the UE0173 I2C D-Pad Module was not present in the selected source package. The top and bottom views in Section 6 show the visible board labels, but they do not establish connector numbering, voltage limits, or signal direction. Verify these details against the product-specific schematic before connecting a cable or designing a carrier.

5 Board Operation

5.1 Before Connecting

Check the product-specific UE0173 pinout or schematic for power, ground, SDA, and SCL. The board views alone do not establish voltage limits or connector numbering.

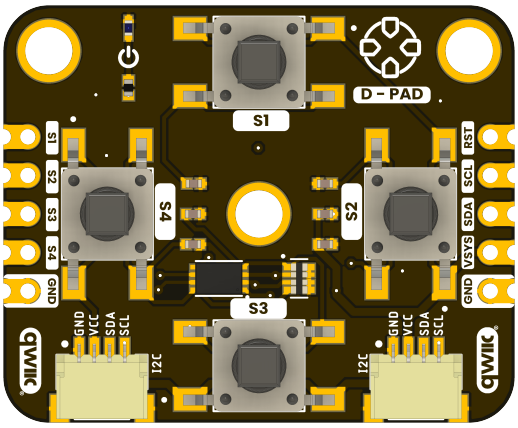
5.2 I2C Bus Check

After confirming the wiring and supply limits, use the generic Arduino or MicroPython I2C scanner in `software/examples/i2c/`. Set the host SDA and SCL pins for your controller. A responding address confirms only that a device acknowledged on the bus; it does not identify the module or validate its measurement protocol.

6 Mechanical Information

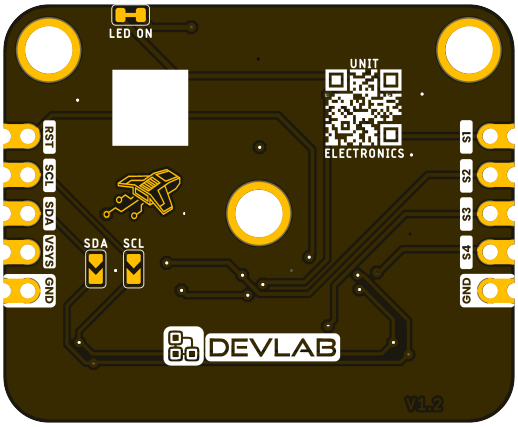
The supplied board views identify the component and underside layouts of the UE0173 I2C D-Pad Module. Use the labeled drawings for fit checks; do not infer dimensions by scaling a render.

6.1 Top View



UE0173 I2C D-Pad Module top view.

6.2 Bottom View



UE0173 I2C D-Pad Module bottom view.

6.3 Dimensional Data

A controlled dimensioned drawing is not available in the selected source package. Verify the board outline, mounting pattern, component heights, and connector clearances before designing an enclosure or carrier.

7 Company Information

Item	Information
Company / brand	UNIT Electronics
Board ecosystem	DevLab
Product family	Atom
Website	UNIT Electronics
Address	Salvador 19, Cuauhtémoc, 06000 Mexico City, CDMX, Mexico

The product repository and controlled technical references are listed in the next chapter.

8 Reference Documentation

Reference	Location
Product repository	UE0173 I2C D-Pad Module
Top view	3D board view
Bottom view	3D board view

A product-specific pinout drawing was not included in the selected source package.

9 Appendix

9.1 Schematic Status

A product-specific schematic was not included in the selected source package.

9.2 Document Control

Item	Value
Product	DevLab I2C D-Pad Module
MPN	UE0173
Source	UNIT Electronics selected product source package

9.3 Data Requiring Confirmation

Electrical limits, I2C commands, and mechanical tolerances must be verified against product-specific released documentation.